

PROJECT DOCUMENTATION

Auto Ticket Classification

using ServiceNow Flow Designer

Platform: ServiceNow (Personal Developer Instance) | Tool: Workflow Studio / Flow Designer

Use case: School IT Ticket Management

1. Introduction

In a School IT help desk, students and staff raise tickets for problems such as Wi-Fi not working, projector issues, forgotten passwords or slow computers. Sorting each ticket into the right category by hand is slow and error-prone.

This project uses ServiceNow Flow Designer to classify tickets automatically. As soon as a new ticket is created without a category, the flow reads the Short Description, decides the correct Category and Subcategory, updates the record, and sends a confirmation email to the caller.

1.1 Objectives

- Create a custom ticket table with all required fields, data types and choices.
- Make Subcategory depend on the selected Category so data stays consistent.
- Build a no-code flow that classifies tickets from keywords in the Short Description.
- Notify the caller by email once the ticket is submitted.
- Test the flow end to end and confirm results in the execution details.

1.2 Tools Used

Tool / Module	Purpose
ServiceNow Developer Instance	Platform where the table, form and flow are built
System Definition > Tables	Create the custom ticket table
Form Design	Add fields to the ticket form using drag and drop
Dictionary Entry	Configure the dependent field (Subcategory on Category)
Workflow Studio (Flow Designer)	Build, test and activate the automation flow

2. Solution Overview

The solution has two parts: (A) the data model, which is the table and its form, and (B) the automation, which is the flow that reacts when a record is created.

Stage	What happens
1. Trigger	A new record is created in the Incident WorkFlow table and Category is empty.
2. Condition check	The flow checks the Short Description for keywords (for example Wi-Fi or Network, Performance).
3. Update Record	The matching branch updates the ticket with the correct classification.
4. Send Email	An email "Your Request for the issue has been submitted" is sent to the caller.

3. Part A: Build the Ticket Table

3.1 Open the required modules

Log in to the ServiceNow instance. Use the All menu filter to open the modules needed. The video begins by opening Local Update Sets, so that all configuration is captured inside an update set.

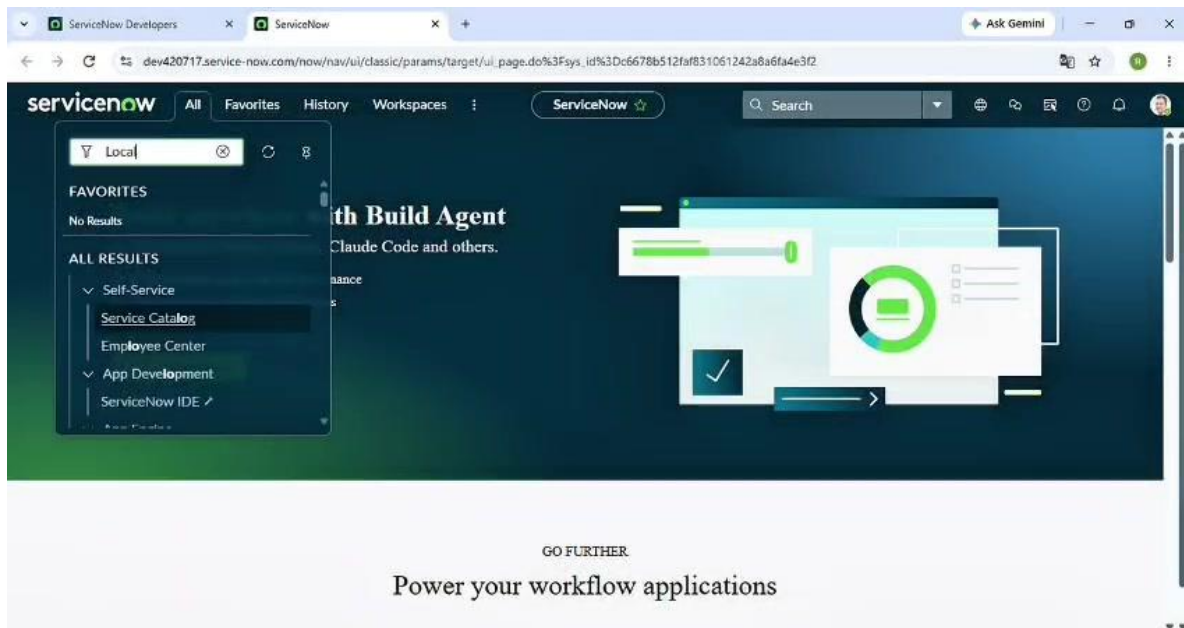


Figure 1: Filter navigator to open System Update Sets > Local Update Sets

3.2 Create the table

1. Go to System Definition > Tables and click New.
2. Enter Label as "Incident WorkFlow"; the Name field is generated automatically (u_incident_workflow).
3. Keep Application as Global and leave "Create module" and "Create mobile module" ticked.
4. Click Submit.

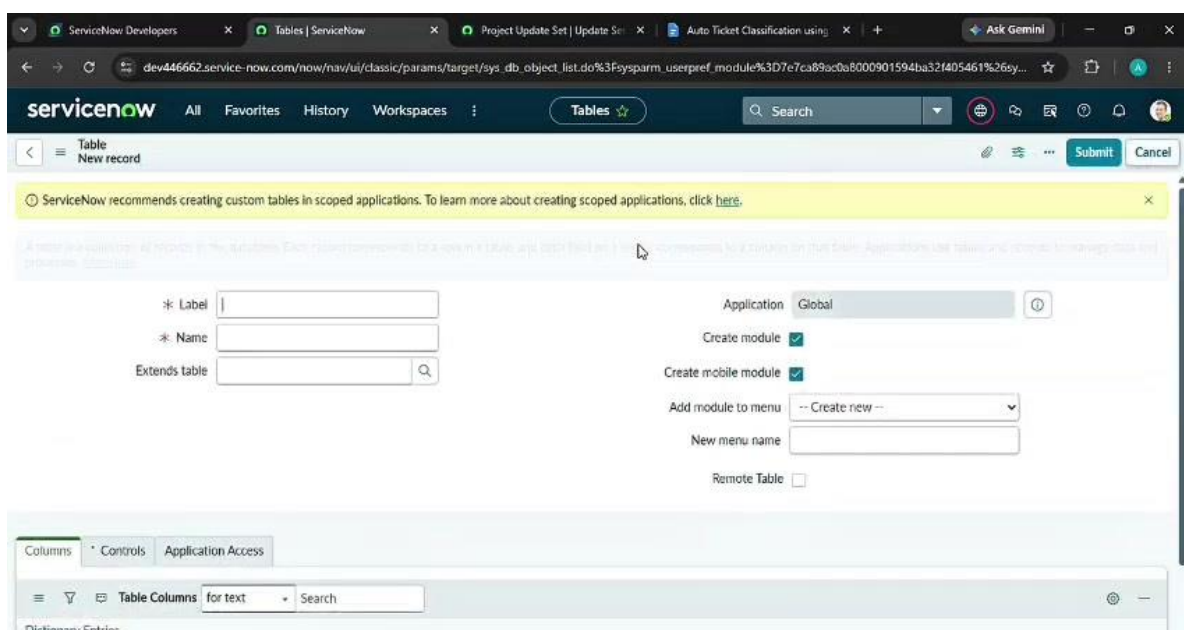


Figure 2: New table record with the label "Incident WorkFlow"

3.3 Configure auto-numbering

Open the Controls tab of the table and set up auto-numbering so every ticket gets a unique ID.

Setting	Value
Prefix	INC
Number (starting)	1,000
Number of digits	7
Create access controls	Ticked
User role	u_incident_workflow_user

The screenshot shows the ServiceNow 'Table - New Record' form, specifically the 'Controls' tab. The form is for creating a new table. The 'Prefix' is set to 'INC', 'Number (starting)' is '1,000', and 'Number of digits' is '7'. The 'Create access controls' checkbox is checked, and the 'User role' is set to 'u_incident_workflow_user'. The 'Auto-number' checkbox is also checked. The form includes a 'Submit' button and a 'Cancel' button.

Figure 3: Controls tab showing prefix INC, 7 digits and the user role

4. Part B: Design the Form and Fields

4.1 Field specification

Fields were planned in a separate Google Doc titled "Auto Ticket Classification using Flow Designer" before being built. The milestone is to create all required fields with correct data types, choices and dependencies.

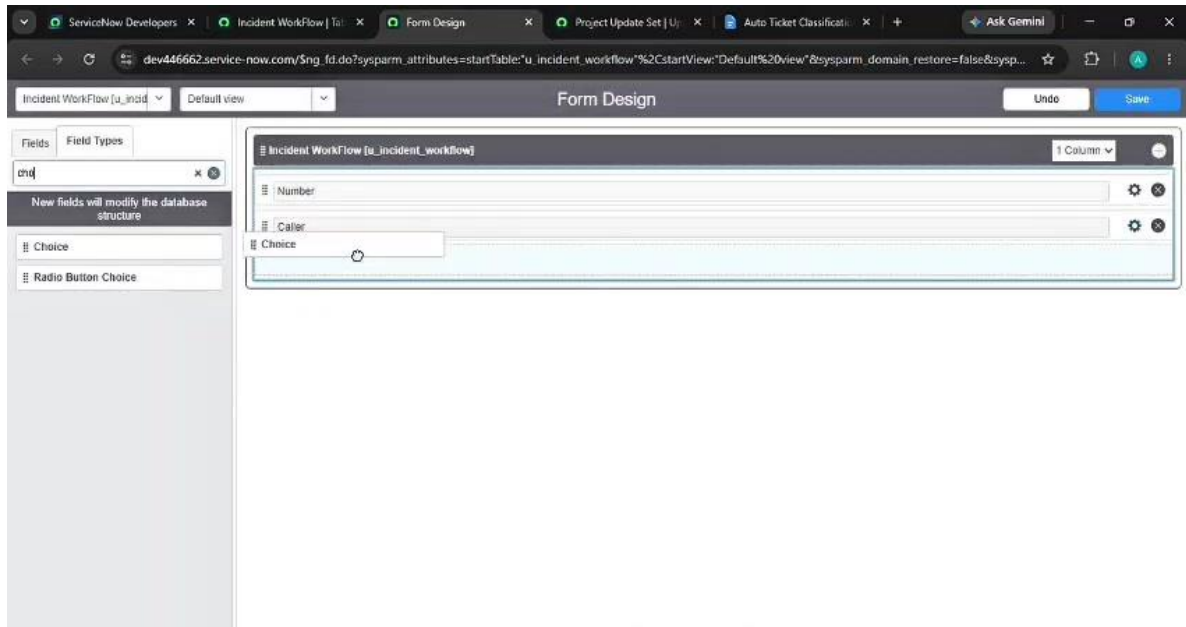


Figure 4: Field specification prepared in Google Docs

Field Label	Type	Reference / Choices
Number	Auto Number	Generated (INC prefix)
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	-
Description	String	-
State	Choice	New, In Progress, On Hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

4.2 Add fields using Form Design

Open the table form and use Form Design. Search the Field Types panel, drag a type onto the form, then set its properties.

1. Drag a Reference field, set the Label to "Caller" and choose the referenced table (User).
2. Drag Choice fields for Category, Subcategory and State, then add their choice values.
3. Drag String fields for Short Description and Description.
4. Drag Reference fields for Assigned Group (Group) and Assigned to (User).
5. Click Save.

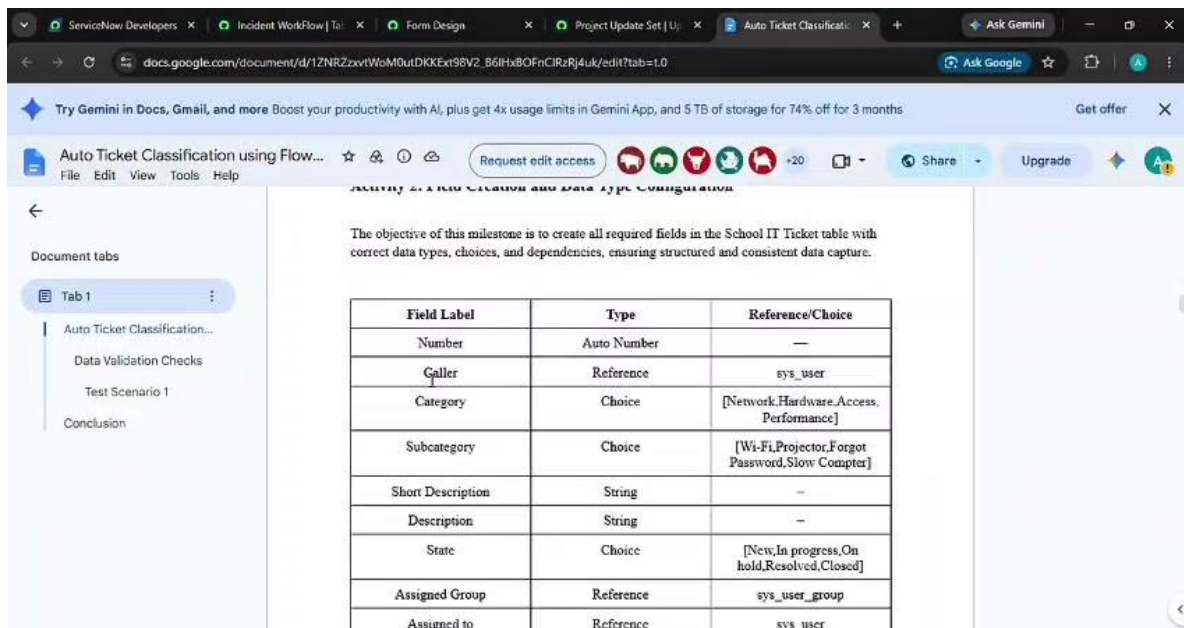


Figure 5: Adding a Reference field and editing its properties

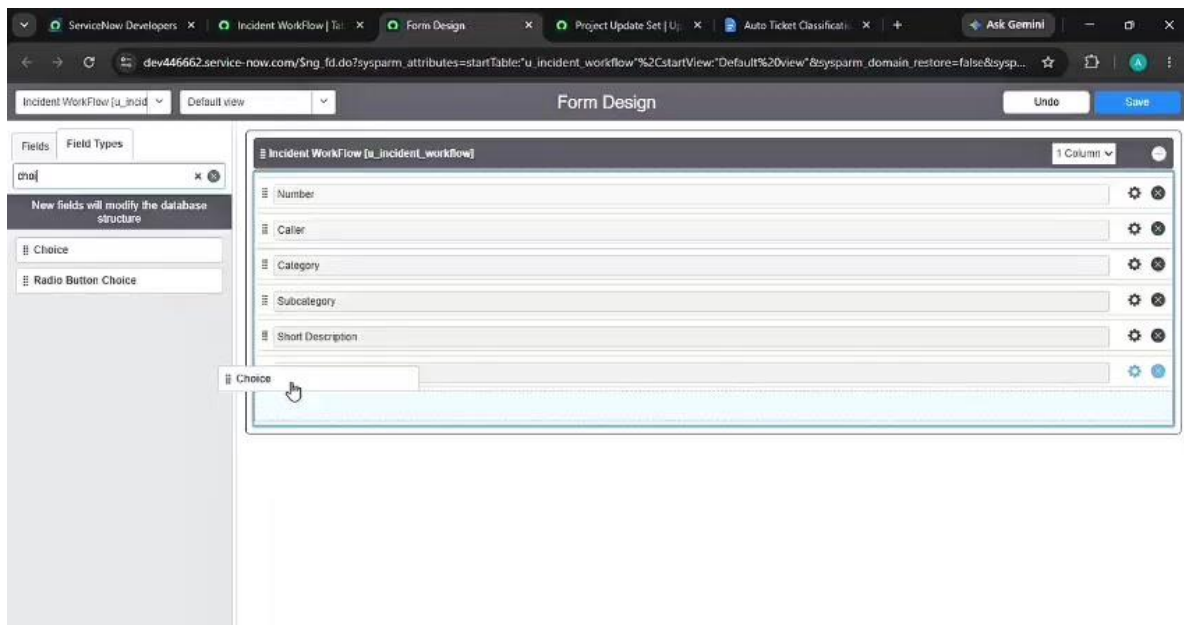


Figure 6: Adding the State choice field

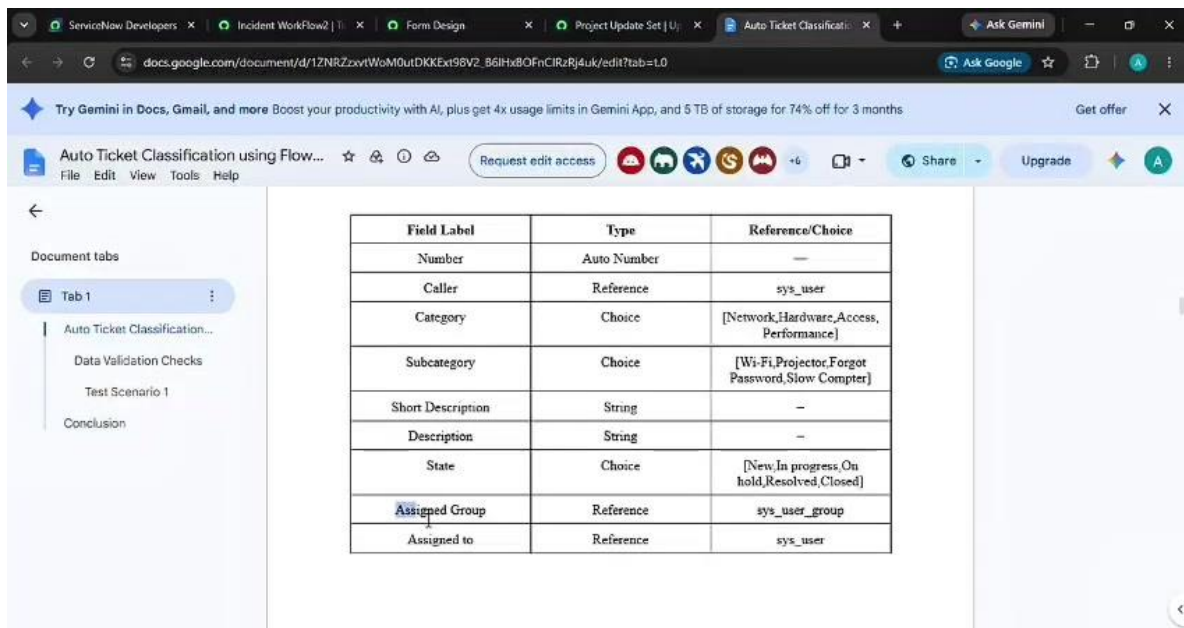


Figure 7: Selecting the referenced table for Assigned Group

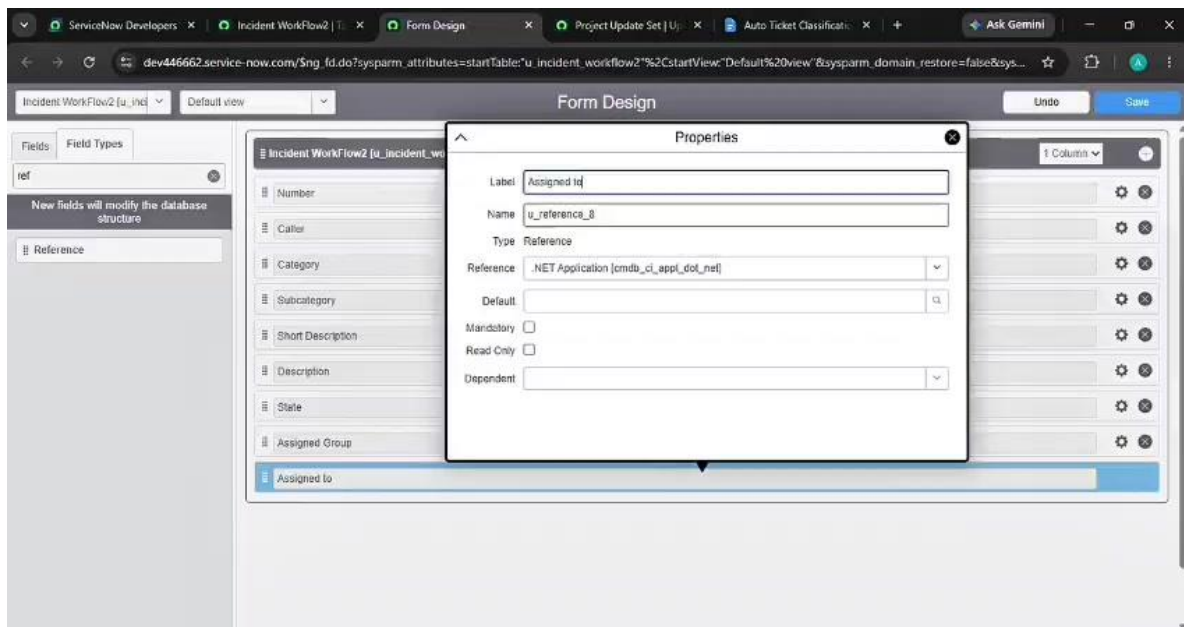


Figure 8: Completed form layout with all nine fields

4.3 Preview the form

Create a new record from the table to check the form. Number is auto-filled (INC0001001) and the fields appear in the designed order.

Table - Incident Workflow2

Create access controls ☒

* User role

Delete Update Delete All Records

Related Links

- [Form Builder](#)
- [Design Form](#)
- [Open a new empty form for this table](#)
- [Show Form](#)
- [Show List](#)
- [Show Schema Map](#)
- [Add to Service Catalog](#)
- [Run Point Scan](#)
- [Explore REST API](#)

Access Controls (4) Security Data Filters Labels (1) Database Indexes (1) Table Subscription Configuration (1)

Access Controls

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_incident_workflow2	Allow If	create	record	true	admin	2026-09-25 06:48:50

Figure 9: New record form showing auto-generated number INC0001001

5. Part C: Make Subcategory Depend on Category

So that users only see subcategories relevant to the chosen category, open the Subcategory dictionary entry (right-click the field label > Configure Dictionary).

1. Open the Dependent Field tab.
2. Tick "Use dependent field".
3. Set "Dependent on field" to Category.
4. Save (right-click header > Save) and add each choice with its dependent value.

Dictionary Entry - Subcategory View: Advanced

* Column name Mandatory ☐

* Max length Display ☐

Read only option

Attributes

Choice List Specification Dependent Field Calculated Value Default Value

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field

Create Choice List Delete Column Update

Figure 10: Dependent Field tab: Subcategory depends on Category

Test on a new record: after selecting Category = Network, the Subcategory list is filtered accordingly.

Dictionary Entry
Subcategory View: Advanced*

Related Links
Show Table
Run Point Scan

Access Controls Choices (4) Attributes (1) Labels (1)

Label Search

Choices

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Slow Computer	slow computer	en	3	false	2026-09-25 07:13:19	
Projector	projector	en	1	false	2026-09-25 07:13:19	
Wi-Fi	wi-fi	en	0	false	2026-09-25 07:13:19	Network
Forgot Password	forgot password	en	2	false	2026-09-25 07:13:19	

1 to 4 of 4

Figure 11: Testing the form: Category set to Network

6. Part D: Build the Flow in Flow Designer

6.1 Create the flow

1. In the All menu, search "flow" and open Process Automation > Flow Designer (Workflow Studio).
2. Click New > Flow.
3. Flow name: "Auto Classify School IT Tickets"; Application: Global.
4. Click Build flow.

Incident Workflow2 - Create INC0001002

Number: INC0001002

Caller:

Category: Hardware

Subcategory: -- None --

Short Description:

Description:

State: -- None --

Assigned Group:

Assigned to:

Submit

Figure 12: Opening Flow Designer from the All menu

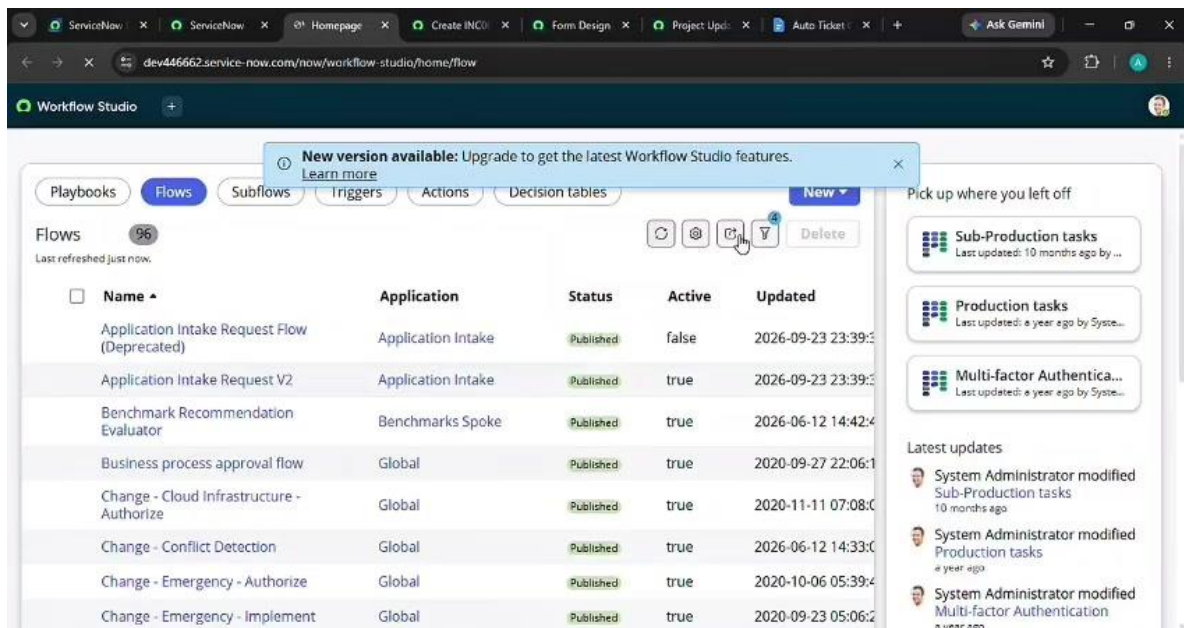


Figure 13: Naming the flow "Auto Classify School IT Tickets"

6.2 Configure the trigger

1. Click Add a trigger and choose Record > Created.
2. Table: Incident WorkFlow [u_incident_workflow].
3. Add a condition: Category is empty, so only unclassified tickets are processed.
4. Click Done.

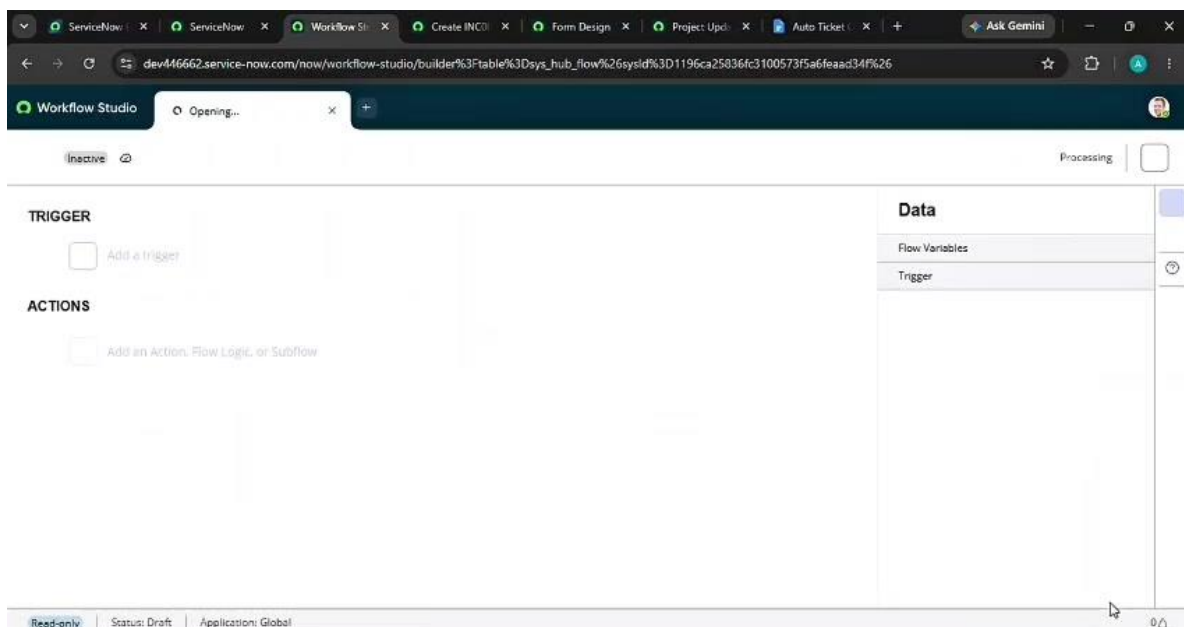


Figure 14: Choosing the Record trigger

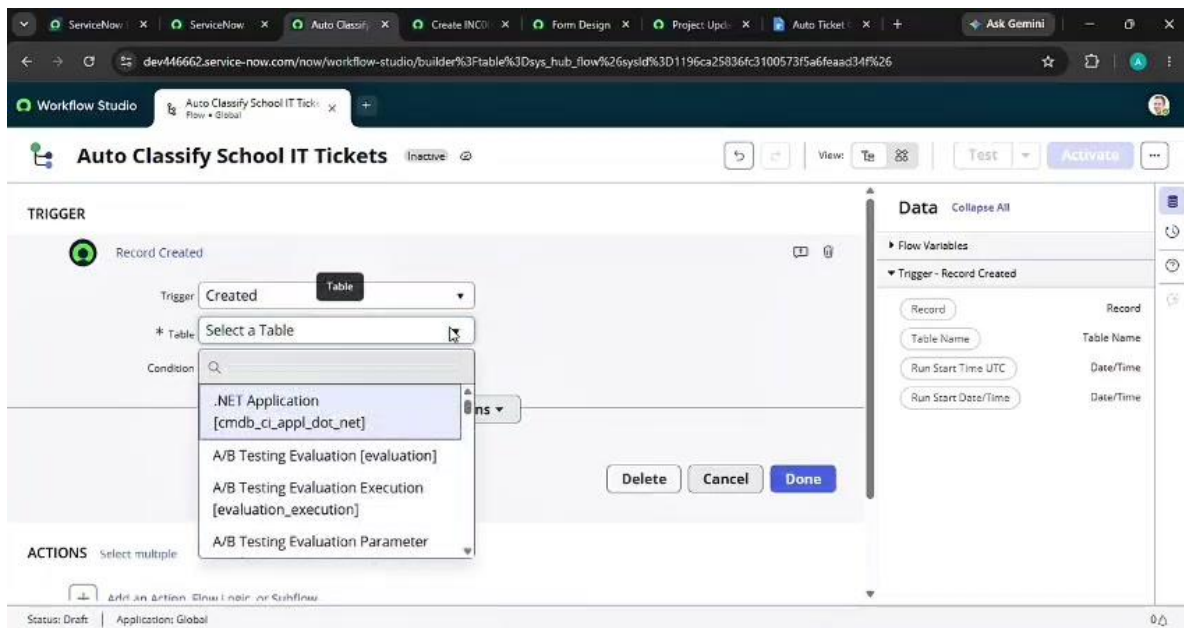


Figure 15: Trigger set to Created on the Incident WorkFlow table

6.3 Add the classification logic

Under Actions, click Add an Action, Flow Logic or Subflow, choose Flow Logic > If. Build one branch per ticket type.

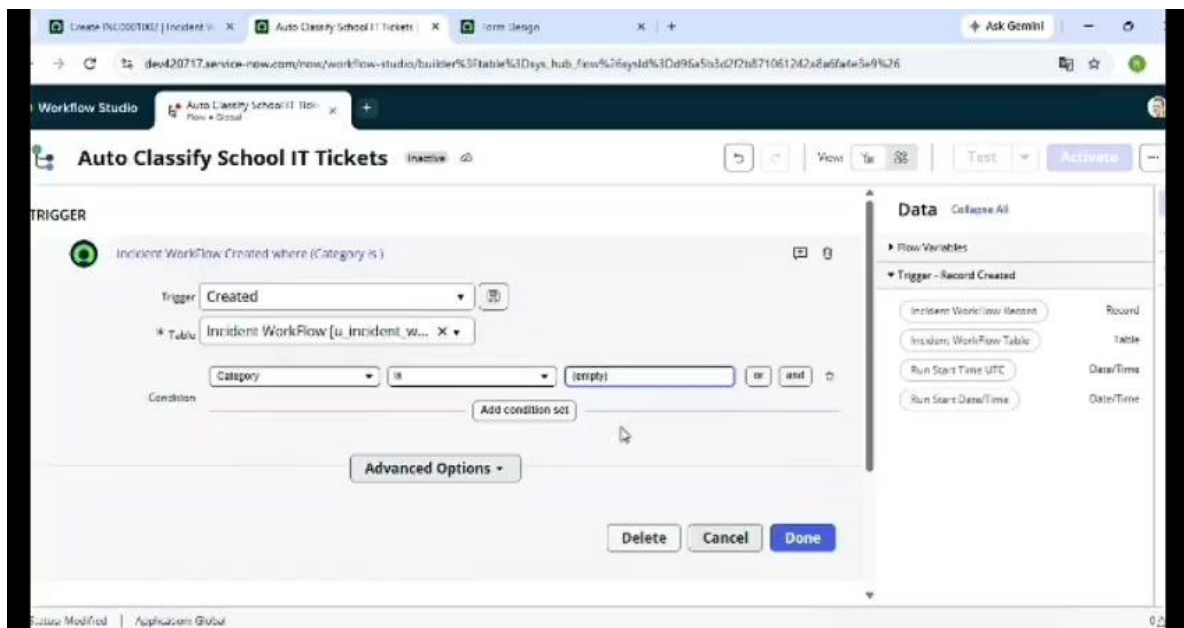


Figure 16: Adding Flow Logic > If

Branch 1 uses the condition label "SD is Wifi or Network". Drag the Short Description data pill from the trigger into each condition:

Condition	Operator	Value
Condition 1: Trigger > Short Description	is	Wi-Fi
OR Condition 2: Trigger > Short Description	is	Network

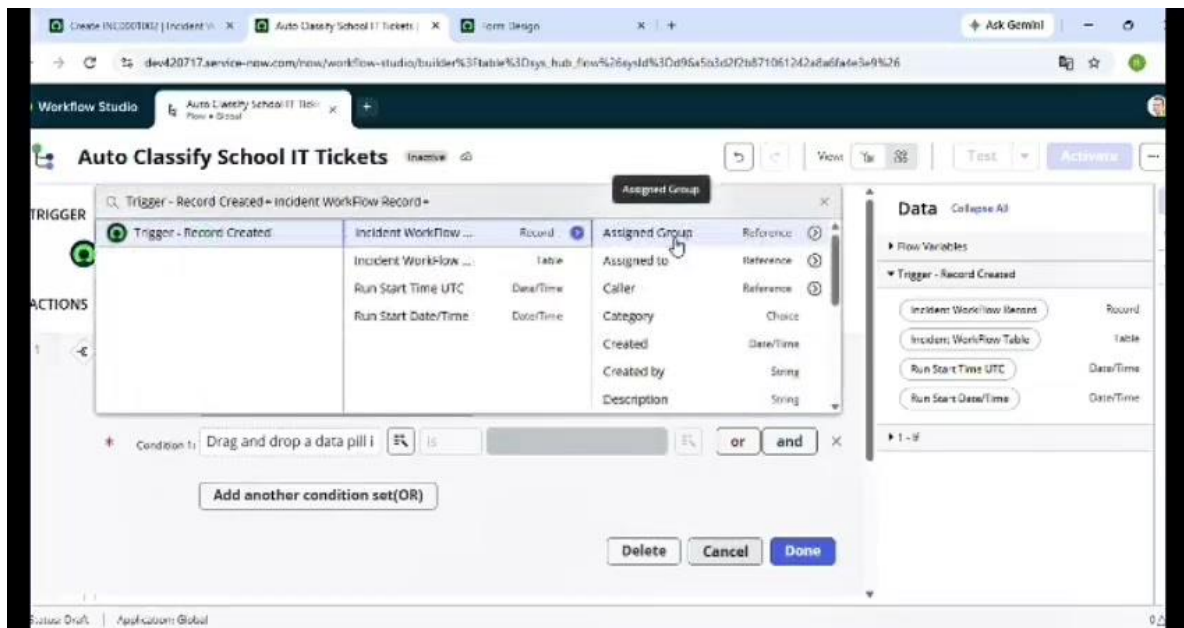


Figure 17: If condition "SD is Wifi or Network" with two OR condition sets

6.4 Update the record

Inside "then", add the Update Record action, drag the trigger record into the Record input, and set the Category and Subcategory values for that branch. Add further "Else If" branches in the same way; the video adds a branch labelled "SD is Performance" with its own Update Record action.

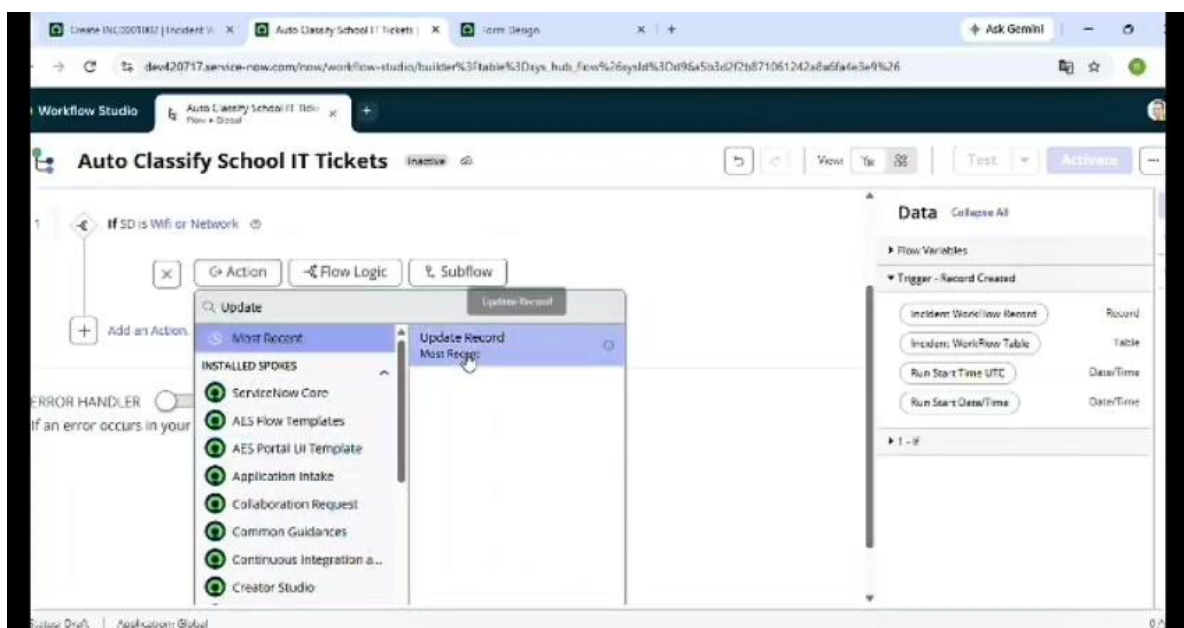


Figure 18: Selecting the Update Record action

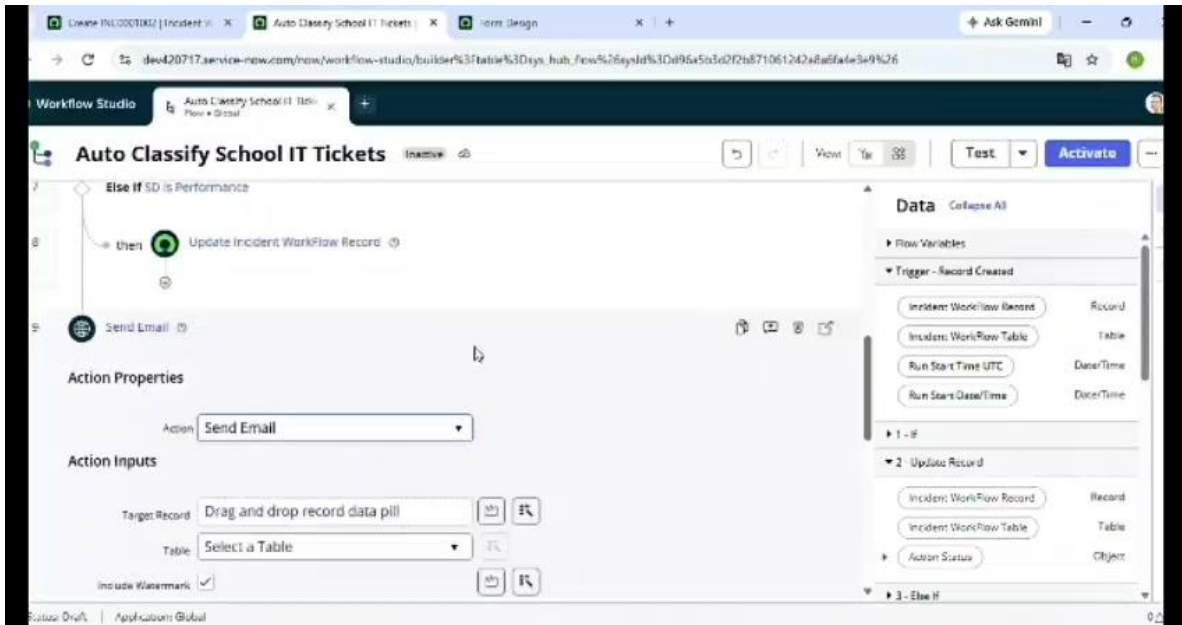


Figure 19: Second branch: Else If "SD is Performance" with Update Record

6.5 Send confirmation email

At the end of the flow add the Send Email action and configure it as follows:

Input	Value
Target Record	Trigger > Incident WorkFlow Record
Table	Incident WorkFlow [u_incident_workflow]
To	Trigger record > Caller > Email
Subject	Your Request for the issue has been submitted

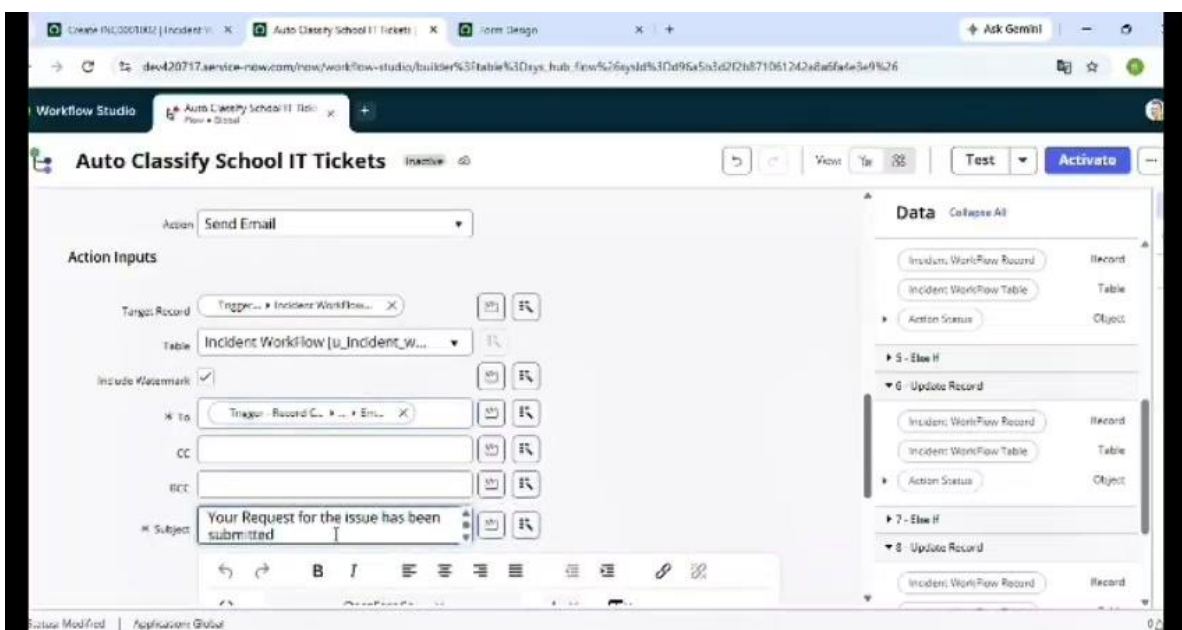


Figure 20: Send Email action with subject entered

6.6 Activate the flow

Click Activate and confirm the prompt "Are you sure you want to activate the flow?". The flow now runs whenever the trigger conditions are met.

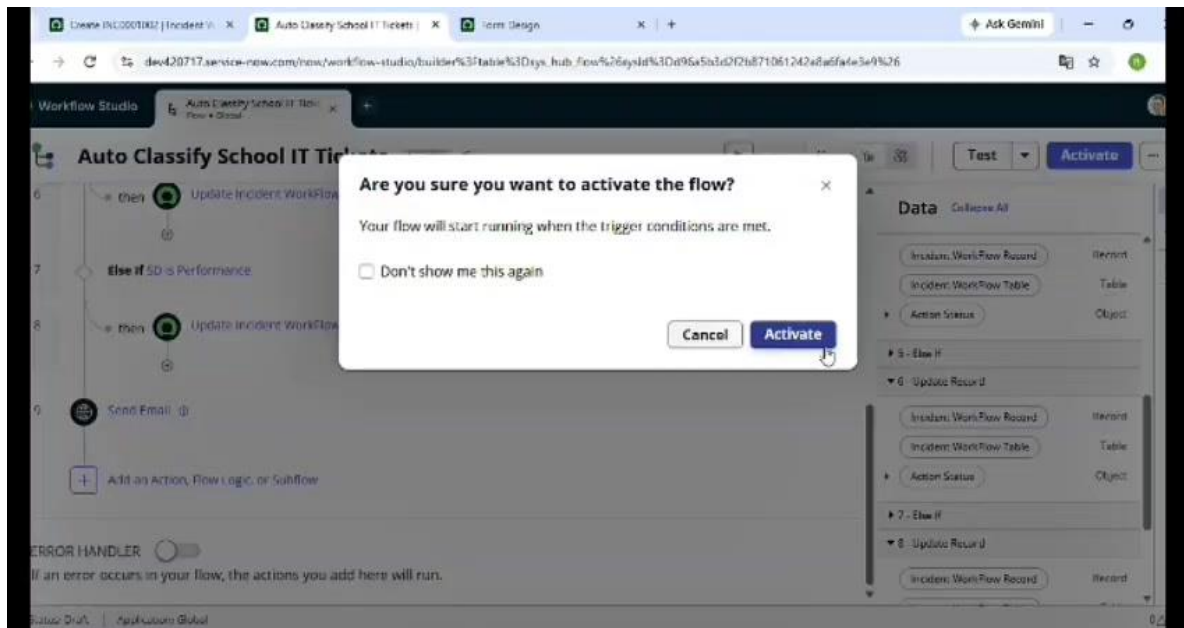


Figure 21: Activation confirmation dialog

7. Part E: Testing

7.1 Test by creating a ticket

1. Open a new record in the Incident Workflow table.
2. Select a Caller (for example, Abel Tuter).
3. Leave Category empty and type Short Description: "WiFi not working in library".
4. Click Submit and reopen the record to check that Category and Subcategory were filled in.

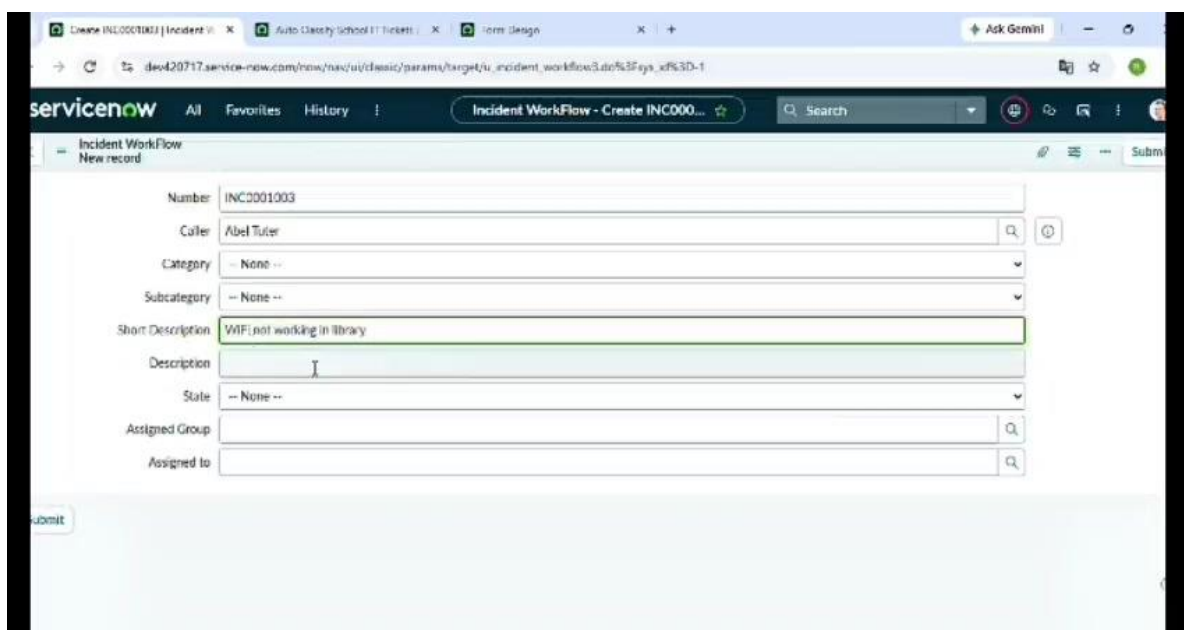


Figure 22: Test ticket with Short Description "WiFi not working in library"

7.2 Use the built-in Test option

In Workflow Studio click Test, pick an existing record (for example INC0001004) and click Run Test. Flow Designer executes every action against that record.

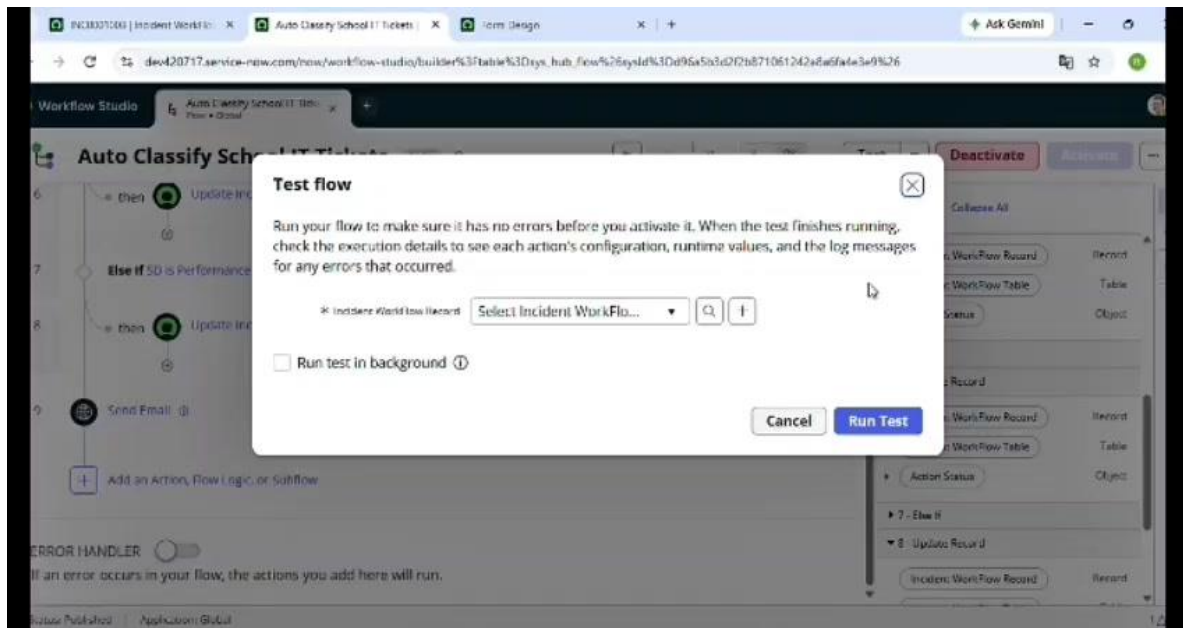


Figure 23: Test flow dialog running against an existing record

7.3 Verify the execution details

Open the Execution Details page. The trigger shows as completed, the matching If branch shows "Evaluated - True" and its Update Record action shows "Completed", while branches that did not match show "Not Run". This confirms the correct classification path was followed.

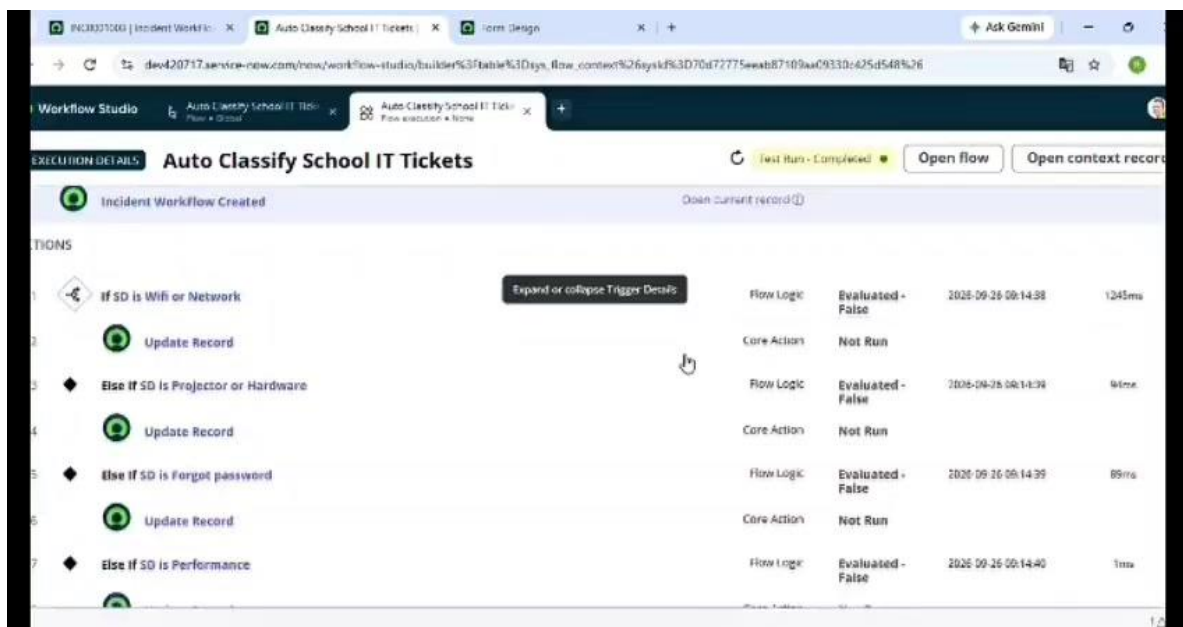


Figure 24: Execution details showing the matched branch evaluated and the others not run

8. Results and Conclusion

- Tickets created without a category are classified automatically from the Short Description.
- Dependent choice lists keep Category and Subcategory consistent.

- The caller receives an email confirmation as soon as the ticket is submitted.
- Execution Details give a clear audit trail of which branch ran and which did not.

Overall, Flow Designer removes manual triage effort from the School IT help desk, reduces mistakes and speeds up the first response, all without writing code.

8.1 Possible Enhancements

- Add more keyword branches (Projector, Password reset, Hardware).
- Auto-assign the Assigned Group based on the category.
- Set State to In Progress once a technician is assigned, and email on Resolved.